Vector Databases for RAG

A vector database is a system built to store embeddings and search them efficiently by similarity, rather than by exact matches like a traditional database. Each item, such as a chunk of text, is represented as a vector: a list of numbers produced by an embedding model that captures its meaning.

In a Retrieval-Augmented Generation system, the vector database is what makes step 5, similarity search, fast enough to use in real time. Instead of comparing a query against every stored chunk one by one, the database uses specialized indexing structures to narrow the search down to a small set of likely candidates almost instantly, even across millions of vectors.

Most vector databases also store the original text and metadata alongside each vector, so that once the closest matches are found, the system can return the actual passage of text, not just a list of numbers.

Similarity Metrics and Indexing

To decide how close two vectors are, a vector database uses a distance metric. Cosine similarity measures the angle between two vectors and ignores their length, which works well for text embeddings where direction matters more than magnitude. Euclidean distance measures straight-line distance between two points. Dot product combines both angle and magnitude, and is often used when embeddings are already normalized.

Comparing a query against every single vector, known as a brute-force or flat search, is accurate but becomes slow as a collection grows into the millions. Approximate Nearest Neighbor, or ANN, indexing algorithms trade a small amount of accuracy for a large speed improvement.

HNSW, short for Hierarchical Navigable Small World, builds a layered graph connecting similar vectors, and searches by hopping across the graph toward the query. IVF, short for Inverted File Index, first groups vectors into clusters, then only searches within the clusters closest to the query. Both are common choices in production vector databases.

Choosing a Vector Database for RAG

Chroma is a lightweight, open-source vector database that can run embedded directly inside a Python application and persist to a local folder on disk, which makes it a good fit for learning projects and small to medium deployments without needing a separate server.

FAISS, built by Meta, is a library rather than a full database. It is very fast and widely used as the indexing engine inside other tools, but on its own it does not handle metadata storage, persistence, or updates as conveniently as a dedicated database.

Pinecone and Weaviate are managed, cloud-hosted vector databases designed for production workloads at scale, offering built-in replication, filtering, and horizontal scaling, at the cost of running as an external hosted service rather than an embedded library.

For a RAG system, the right choice usually depends on scale: an embedded database like Chroma is often enough for a single application with a modest amount of data, while a managed service becomes more attractive once multiple services need to share the same index or the dataset grows very large.